Kernel Functions

$\Phi(x) = x$

① Linear kernel: $k(x_i, x_j) = x_i^T . x_j$

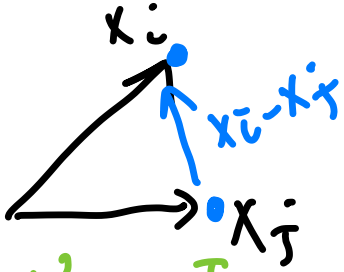
② Polynomial kernel: $k(x_i, x_j) = (x_i^T . x_j + 1)^9$ — 9th order polynomial

③ Sigmoidal kernel: $k(x_i, x_j) = \tanh(2 . x_i^T . x_j + 1)$

④ Gaussian kernel: $k(x_i, x_j) = \exp\left[-\frac{\|x_i - x_j\|_2^2}{2s^2}\right]$ — ∞th order polynomial

d_{ij}^2

$[0, 1]$



$\|w\|_2^2 = w^T . w$

$$\|x_i - x_j\|_2^2 = (x_i - x_j)^T (x_i - x_j)$$

$$= x_i^T . x_i - 2\, x_i^T . x_j + x_j^T . x_j$$

maximize $\sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j k(x_i, x_j)$

subject to: $\sum_{i=1}^{N} \alpha_i y_i = 0$

$0 \leq \alpha_i \leq C \quad \forall i$

$\begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{bmatrix}$

$\mathbf{1}^T . \alpha$

$\begin{bmatrix} y_1 & y_2 & \dots & y_N \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{bmatrix} = 0$

$y^T . \alpha = 0$

$0 \leq \alpha_1 \leq C$

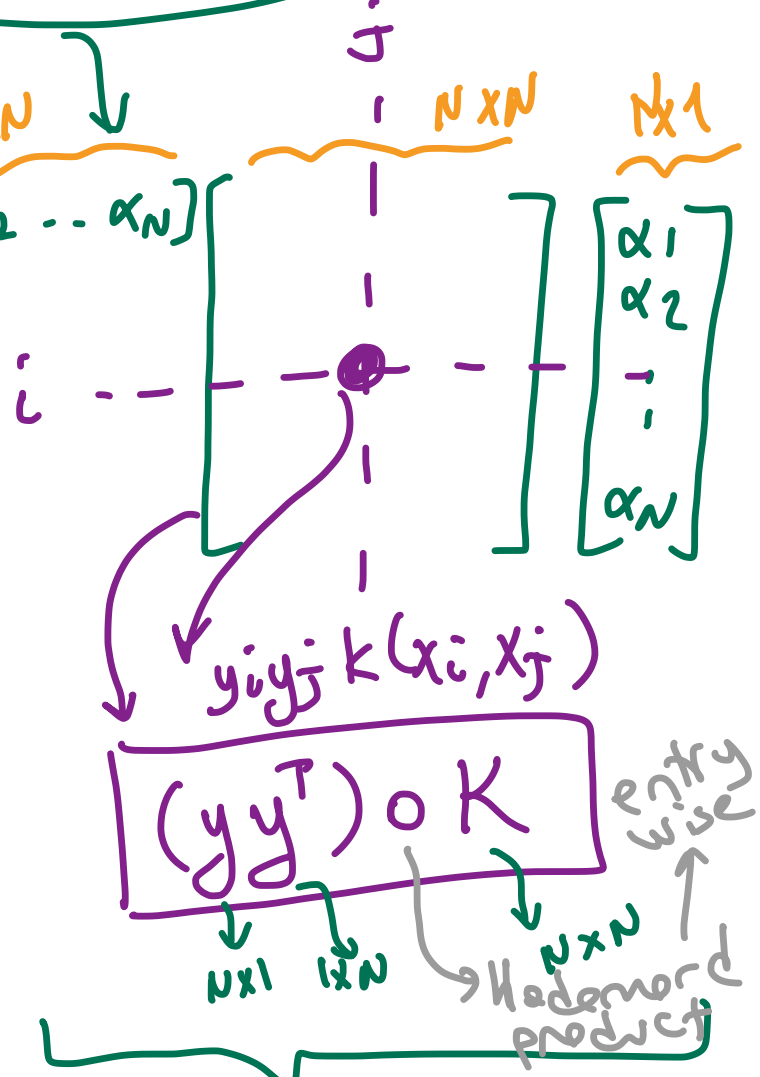
$0 \leq \alpha_2 \leq C$

$\vdots$

$0 \leq \alpha_N \leq C$

$0 \leq \alpha \leq \mathbf{1} . C$

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 12 \end{bmatrix}$



$-\frac{1}{2} . \alpha^T . ((yy^T) \circ K) \alpha$

Maximize $1^T\alpha - \frac{1}{2}\alpha^T((yy^T)\circ K)\alpha$

Subject to: $y^T.\alpha = 0$

$0 \leq \alpha \leq C.1$

K should be a p.s.d. (positive semidefinite) matrix. (all of its eigenvalues are nonnegative)

$\alpha^T.K.\alpha \geq 0 \quad \forall \alpha \in \mathbb{R}^N$

$\alpha^T.K.\alpha > 0 \quad \forall \alpha \in \mathbb{R}^N$ except $\alpha = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ (positive definite)

Constructing Kernels

$k(x_i, x_j)$ a valid kernel $\Rightarrow$ $c.k(x_i, x_j)$ a valid kernel ($c \geq 0$ (nonnegative))

Proof: we know that $\alpha^T.K.\alpha \geq 0$,

$\alpha^T.((cK).\alpha \geq 0$?

$c.\alpha^T.K.\alpha \geq 0$

($c \geq 0$, $\alpha^T.K.\alpha \geq 0$)

$k_1(x_i, x_j)$

$k_2(x_i, x_j)$

$>$ $c_1 \cdot k_1(x_i, x_j) + c_2 k_2(x_i, x_j)$ is also a valid kernel.

$c_1 \geq 0$ $\quad$ $c_2 \geq 0$

Exercise: show that $k_1(x_i, x_j) \cdot k_2(x_i, x_j)$ is also a valid kernel.

Multiclass kernel machines

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in \mathbb{R}^D$, $y_i \in \{1, 2, \ldots, K\}$

One-versus-all (OVA)

of classifiers $= K$

training set size $= N$

1^+ vs $\{2, 3, \ldots, K\}^-$ $\Rightarrow SVM_1$

2^+ vs $\{1, 3, \ldots, K\}^-$ $\Rightarrow SVM_2$

$\vdots$

K^+ vs $\{1, 2, \ldots, K-1\}^-$ $\Rightarrow SVM_K$



One-versus-other (OVO)

$\#\text{ of classifiers} = \frac{K(K-1)}{2}$

training set size $\cong 2 \cdot \frac{N}{K}$

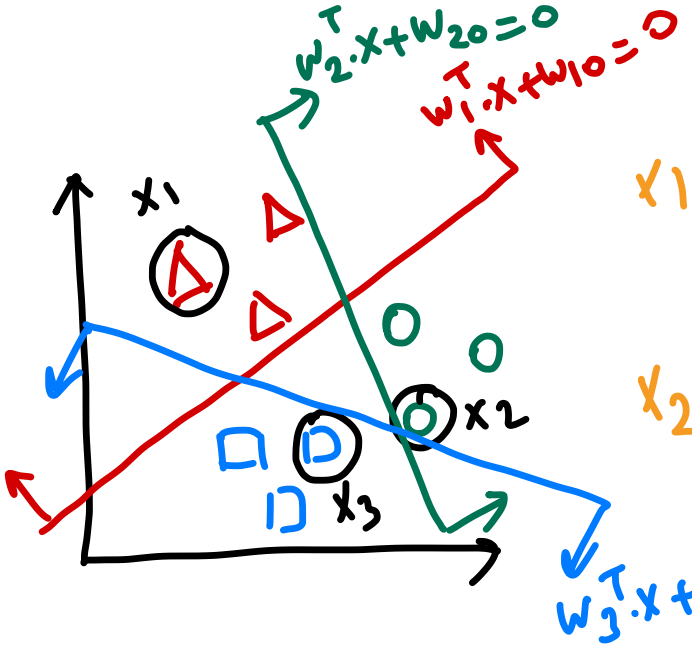
1^+ vs 2^- $\Rightarrow SVM_{1vs2}$

1^+ vs 3^- $\Rightarrow SVM_{1vs3}$

$\vdots$

$(K-1)^+$ vs K^- $\Rightarrow SVM_{Kvs(K-1)}$

x_{N+1} → SVM_{1vs2}, SVM_{1vs3}, - - - , $SVM_{Kvs(K-1)}$ ⇒ pick the one with highest # of wins.



x_1:

$w_1^T x_1 + w_{10} \geq w_2^T \cdot x_1 + w_{20} + 2 - \xi_{12}$

$w_1^T \cdot x_1 + w_{10} \geq w_3^T \cdot x_1 + w_{30} + 2 - \xi_{13}$

x_2:

$w_2^T \cdot x_2 + w_{20} \geq w_1^T \cdot x_2 + w_{10} + 2 - \xi_{21}$

$w_2^T x_2 + w_{20} \geq w_3^T \cdot x_2 + w_{30} + 2 - \xi_{23}$

x_3:

$w_3^T \cdot x_3 + w_{30} \geq w_1^T \cdot x_3 + w_{10} + 2 - \xi_{31}$

$w_3^T \cdot x_3 + w_{30} \geq w_2^T \cdot x_3 + w_{20} + 2 - \xi_{32}$

minimize $\frac{1}{2}\sum_{c=1}^{K}\|w_c\|_2^2 + C\sum_{i=1}^{N}\sum_{c=1}^{K}\xi_{ic}$

subject to: $w_{y_i}^T \cdot x_i + w_{y_i 0} \geq w_c^T \cdot x_i + w_{c0} + 2 - \xi_{ic} \quad \forall (i, c \neq y_i)$

$\xi_{ic} \geq 0 \quad \forall (i, c \neq y_i)$

of decision variables $= K.(D+1) + N(K-1)$

of constraints $= N(K-1)$

Kernel Machines for Regression

$$g(x) = w^T.x + w_0 = \hat{y}$$

ϵ-insensitive loss

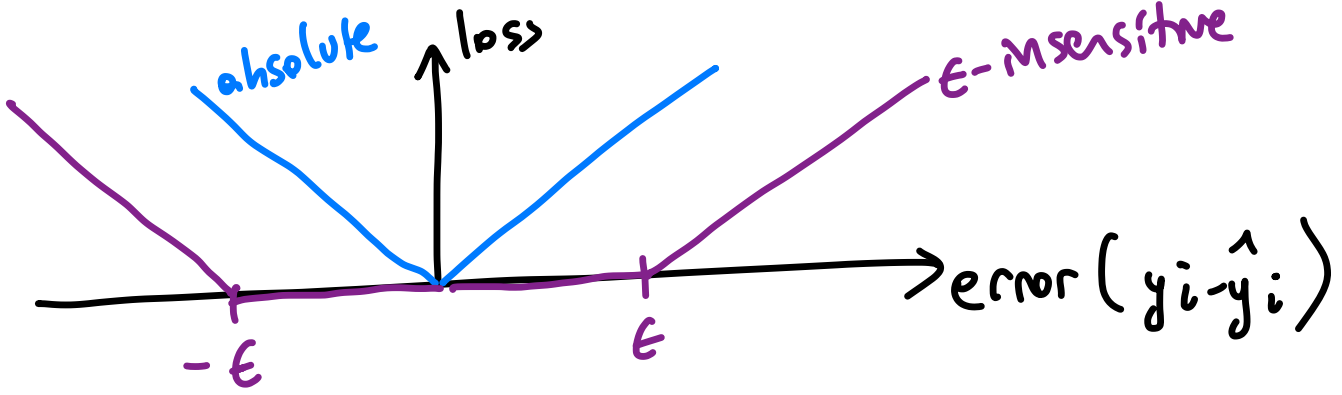
$$\begin{cases} 0 & \text{if } |y_i - \hat{y}_i| \leq \epsilon \\ |y_i - \hat{y}_i| - \epsilon & \text{otherwise} \end{cases}$$

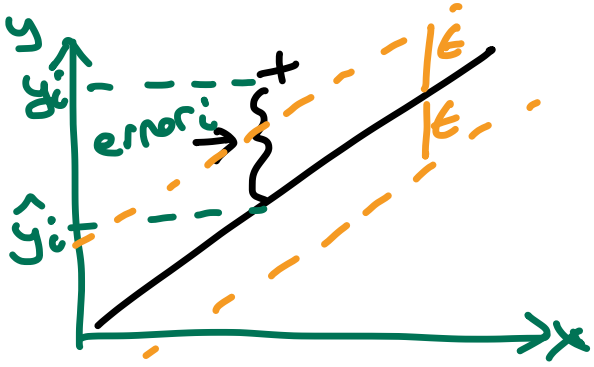
least squared regression

loss $\Rightarrow \sum_{i=1}^{N}(y_i - \hat{y}_i)^2$

very sensitive to outliers







minimize $\frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{N}(\xi_i^+ + \xi_i^-)$

subject to:

underestimation ← $y_i - [w^T \cdot x_i + w_0] \le \epsilon + \xi_i^+ \quad \forall i$

overestimation → $[w^T \cdot x_i + w_0] - y_i \le \epsilon + \xi_i^- \quad \forall i$

$\xi_i^+ \ge 0 \quad \forall i$

$\xi_i^- \ge 0 \quad \forall i$

of decision variables $= (D+1) + 2N$

of constraints $= 2N$

Exercise

maximize $\sum_{i=1}^{N} y_i[\alpha_i^+ - \alpha_i^-] - \epsilon\sum_{i=1}^{N}(\alpha_i^+ + \alpha_i^-) - \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}(\alpha_i^+ - \alpha_i^-)(\alpha_j^+ - \alpha_j^-)x_i^T x_j$

subject to: $\sum_{i=1}^{N}(\alpha_i^+ - \alpha_i^-) = 0$

$0 \le \alpha_i^+ \le C \quad \forall i$

$0 \le \alpha_i^- \le C \quad \forall i$

of decision variables $= 2N$

of constraints $= 1$